

1.3 Exit Note: Unit Conversions

Reference (provided). $1 \text{ mi} = 5280 \text{ ft}$ $1 \text{ m} \approx 3.28 \text{ ft}$ $1 \text{ ft} = 12 \text{ in}$
 $1 \text{ km} = 1000 \text{ m}$ $1 \text{ h} = 3600 \text{ s}$ $1 \text{ min} = 60 \text{ s}$

1. Round each value to three significant figures. Copy the calculator display followed by three dots, then round: $\sqrt{2} = 1.4142135 \dots \approx 1.41$

(a) $\sqrt{3} = 1.7320508076 \dots$

(b) $\frac{100 \text{ m}}{9.58 \text{ s}}$ (Usain Bolt's average speed
in the 100 m world record)

2. Scientific notation.

- (a) One year is 31,556,952 seconds. Write this in scientific notation, rounded to three significant figures.

- (b) A hydrogen atom is 1.06×10^{-10} meters across. Write this as an ordinary decimal number.

3. The Duomo di Milano is 108 meters tall. What is its height in feet? Write the conversion as a fraction and show the units cancelling.

4. A Category 1 hurricane begins at a wind speed of 74 miles per hour. What is this speed in feet per second?

Building	Advertised speed
Shanghai Tower	20.5 meters per second
Taipei 101	1010 meters per minute
One World Trade Center	2000 feet per minute
Empire State Building (express)	1200 feet per minute

- (a) Convert all four speeds to meters per second, rounded to three significant figures. Write each one as a single chain of fractions and show the units cancelling.
- (b) List the four elevators from slowest to fastest. Is that ranking supported by the precision of the data? Explain.
- (c) Our classroom is 18 meters above the building entrance. If each of these elevators were installed in our building and ran at its advertised speed, how long would the ride up take? Compare with the 75 seconds the stairs took us.
- (d) A real elevator has to speed up at the start of the ride and slow down at the end. Explain why your answers to part (c) are all too small — and why the error is worst for the Shanghai Tower.